

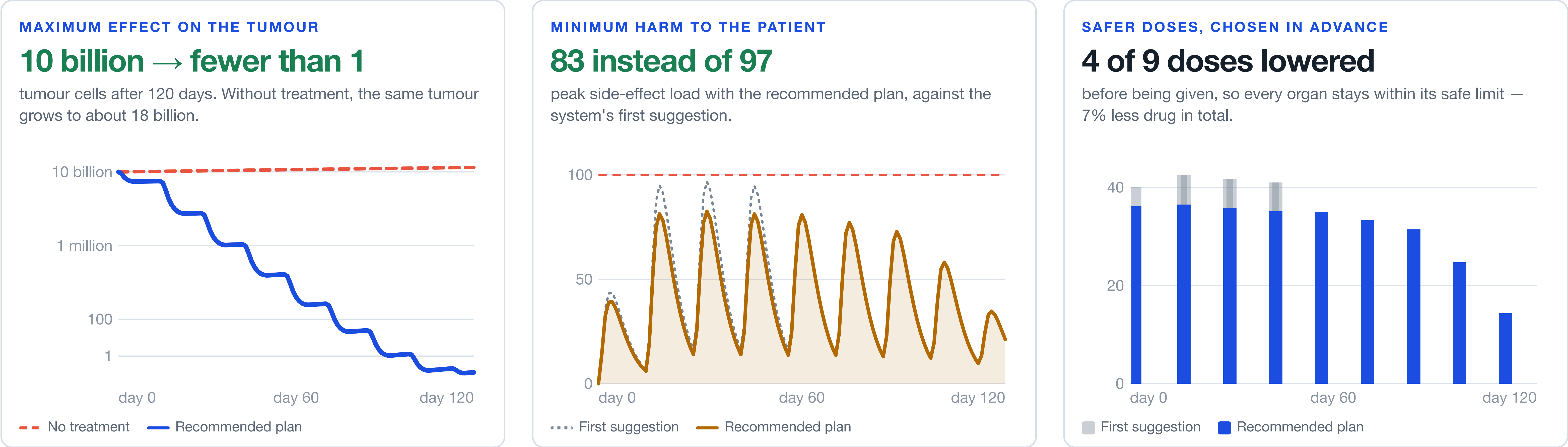
Decision Support System for Chemo

Maximum effect on the tumour, Minimum harm to the patient.

CASE STUDY

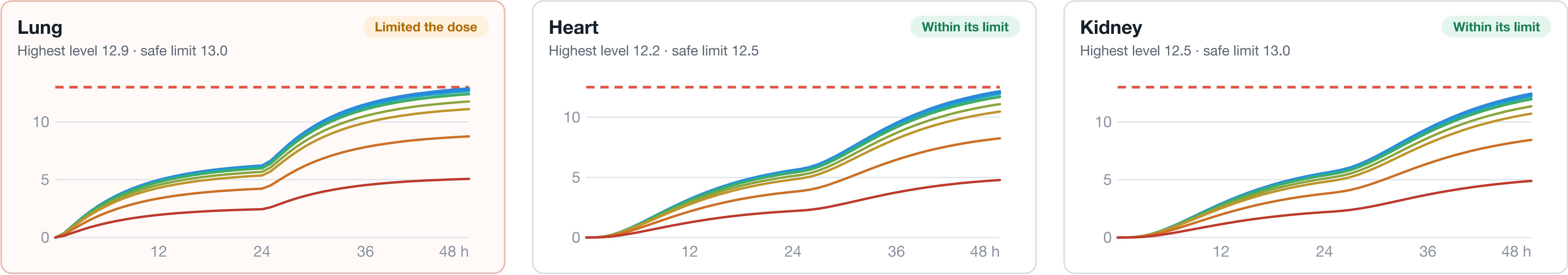
A 70 kg adult with bulky disease, treated for 120 days

About 10 billion tumour cells at the start (~10 g of tumour), a dose every 14 days, and standard adult organ limits. Every figure here is the system's own output for this case.



Drug level in key organs after each dose The red dashed line is each organ's safe limit — every dose stays below it.

Dose 1 Dose 2 Dose 3 Dose 4 Dose 5 Dose 6 Dose 7 Dose 8 Dose 9



WHAT THE NUMBERS MEAN

Tumour cells

How much cancer is left, counted on a logarithmic scale because treatment kills a share of the cells, not a fixed number. This patient starts near **10 billion** cells, about 10 g of tumour.

Side-effect load

How much drug the body has had to cope with: it builds up after each dose and eases between them. It is the **model's own scale, not a clinical grade** — 100 is the reference maximum the plan is kept under.

Safe limit

The most drug an organ should be exposed to in the 48 hours after a dose, set for each of the **13 organs**. A dose that would cross a limit is lowered first.

Dose

The amount given on the cycle day and the day after. Shown in the model's own units, so the figures compare doses with each other rather than with a drug label.

PROBLEM

Chemotherapy has to shrink the tumour without harming healthy organs — but the dose is usually lowered only **after** the patient already has side effects.

SOLUTION

Before a dose is given, the clinician sees what it is expected to do to the tumour and to **each major organ** — and which organ is limiting it.

IMPACT

The full course is reviewed and alternatives compared **before treatment starts**, so the dose is decided in advance rather than in reaction.

Why it matters for doctors

- 📊

Fewer calculation mistakes
Body size and every dose adjustment are worked out automatically, not by hand.
- 👁️

See the impact first
The tumour's response and the side effects are projected over the whole course.
- 🛡️

13 organs checked
A dose that would push any organ past its safe limit is lowered before it is given.
- 👤

Personalised to each patient
Adjusts for body size, tumour burden and organ tolerance.
- 📋

Compare before deciding
Put schedules side by side and choose the best balance for the patient.
- 🧠

Thinks the way a doctor does
Mimics how an oncologist reasons, and explains every dose in plain words.

